Unit

12 Reacting masses

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Calculations involving moles and masses
- Determining the percentage by mass of an element in a compound
- Determining the empirical formula of a compound from experimental data
- Calculations involving chemical equations and reacting masses

12.1 Calculating the amounts of substances

When chemists do experiments, they need to know the amounts of substances involved in chemical reactions. For example, they have to find out how much of each reactant is required to produce a certain amount of the product. Hence it is very important to know the method of calculating the amounts of substances in chemical reactions.

12.2 Counting numbers by weighing

Just as a bank teller counts coins and bills, a chemist counts atoms, molecules and formula units of substances. Unlike a bank teller, a chemist cannot count individual items because the particles of matter are too small and too numerous.

Imagine we have a large pile of identical coins. We are required to count 1 000 coins out of this pile. One method of doing this is to count the coins by hand (Fig. 12.1). We can also do it by weighing if we know the mass of each coin (Fig. 12.2). For example, if the mass of each coin is 1 g, we can just weigh out 1 000 g of coins on a balance. This method is much quicker.



Fig. 12.1 Counting coins by hand



Fig. 12.2 Counting coins by weighing

Consider two piles of coins — A and B (Fig. 12.3). Pile A consists of silver coins each with a mass of 1 g. Pile B consists of gold coins each with a mass of 10 g. Suppose there is an equal number of coins in each pile. The mass of the pile of gold coins is 10 times the mass of the pile of silver coins. This is because the mass of each gold coin is ten times the mass of each silver coin.

Thus, for an equal number of coins, the relative masses of the two different piles are the same as the relative masses of the two different individual coins. In other words, if we weigh out the silver and gold coins according to their relative masses, the number of coins in each pile should be equal.

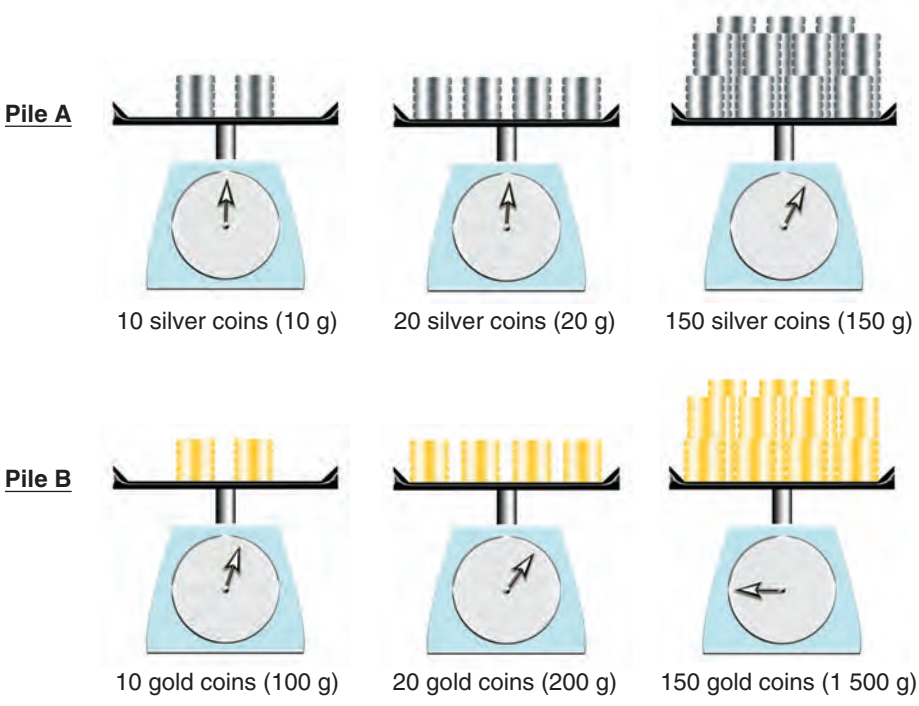


Fig. 12.3 Relationship between the number of coins and their relative masses

Chemists count atoms by weighing in the same way we have considered above. Consider three elements — hydrogen, carbon and phosphorus. Their relative atomic masses are 1.0, 12.0 and 31.0 respectively.

Element	Hydrogen	Carbon	Phosphorus
Relative atomic mass	1.0	12.0	31.0

From the relative atomic masses, we can see that a carbon atom is twelve times as heavy as a hydrogen atom. A phosphorus atom is 31 times as heavy as a hydrogen atom.

Therefore

1.0 g of hydrogen atoms
12.0 g of carbon atoms
31.0 g of phosphorus atoms

contain the same number of atoms.

12.3 What is a mole?

Atoms are very small. It is difficult to count the atoms one by one. So, chemists invented the **mole** as a counting unit, similar to the way a baker uses the word 'dozen', meaning twelve of something (Fig. 12.4).



Fig. 12.4 Eggs are packed in dozens

✓ One mole (symbol: mol) is the amount of a substance that contains the same number of particles as there are atoms in 12.0 g of carbon-12 ($^{12}_{6}C$).

Experiments show that 12.0 g of carbon-12 ($^{12}_{6}C$) contains 6.02×10^{23} atoms. Chemists call this number the **Avogadro constant** (symbol: L). Avogadro was the first scientist to investigate this concept. We can represent the Avogadro constant in the following way:

$L = 6.02 \times 10^{23}\ mol^{-1}$

mole 摩爾　　Avogadro constant 亞佛加德羅常數

Every nitrogen molecule contains two nitrogen atoms. Thus, one mole of nitrogen molecules contains 2 x 6.02 x 10^{23} nitrogen atoms.

One mole of a substance contains 6.02 x 10^{23} particles (Fig. 12.5). For example, one mole of sulphur atoms contains 6.02 x 10^{23} sulphur atoms. One mole of sodium ions contains 6.02 x 10^{23} ions. One mole of nitrogen molecules contains 6.02 x 10^{23} molecules.



Fig. 12.5 1 mole of various substances

We can work out the number of particles in a substance in the following way:

✓ Number of particles in a substance
= number of moles of the substance x L

Example 12.1

Q How many sodium atoms are there in 3.80 moles of sodium atoms?

A Number of sodium atoms
= number of moles of sodium atoms x L
= 3.80 mol x 6.02 x 10^{23} mol^{-1}
= 2.29 x 10^{24}

∴ there are 2.29 x 10^{24} atoms in 3.80 moles of sodium atoms.

Example 12.2

Q How many moles of molecules are there in 2.77×10^{23} oxygen molecules?

A Number of moles of oxygen molecules

$$= \frac{\text{number of oxygen molecules}}{\text{L}}$$

$$= \frac{2.77 \times 10^{23}}{6.02 \times 10^{23}\ \text{mol}^{-1}}$$

$$= 0.460\ \text{mol}$$

$\therefore$ there are 0.460 mole of molecules in 2.77×10^{23} oxygen molecules.

Example 12.3

Q a) How many formula units are there in 4.50 moles of potassium sulphate (K_2SO_4)?

b) How many ions are present?

A a) Number of formula units of K_2SO_4
= number of moles of K_2SO_4 x L
= 4.50 mol x $6.02 \times 10^{23}\ \text{mol}^{-1}$
= 2.71×10^{24}

b) One formula unit of potassium sulphate contains two potassium ions and one sulphate ion.

Number of ions present
= 3 x number of formula units of K_2SO_4
= $3 \times 2.71 \times 10^{24}$
= 8.13×10^{24}

$\therefore$ there are 2.71×10^{24} formula units and 8.13×10^{24} ions in 4.50 moles of potassium sulphate.

Practice 12.1

1 Complete the following table.

Substance	Number of moles of substance present	Number of particles present
Sulphur atoms	0.900 mol	5.42×10^{23} atoms
Zinc ions	6.50 mol	3.91×10^{24} ions

Refer to page T34.

2 a) How many molecules are there in 1.50 moles of carbon dioxide molecules?
9.03×10^{23}

b) How many atoms are present?
2.71×10^{24}

(Avogadro constant = $6.02 \times 10^{23}\ \text{mol}^{-1}$)

12.4 What is molar mass?

The relative atomic mass of hydrogen is 1.0. We have learnt that 1.0 g of hydrogen contains the same number of atoms as 12.0 g of carbon-12. Therefore 1.0 g of hydrogen contains 6.02×10^{23} hydrogen atoms. This means that one mole of hydrogen atoms has a mass of 1.0 g.

The relative atomic mass of phosphorus is 31.0. Since 31.0 g of phosphorus contains 6.02×10^{23} phosphorus atoms, it means that one mole of phosphorus atoms has a mass of 31.0 g.

One mole of atoms of an element has a mass equal to its relative atomic mass expressed in grams.

✓ The mass of one mole of a substance is called its **molar mass**. The units of molar mass are gram per mole ($g\ mol^{-1}$).

Molar mass of a gaseous element

When we talk about one mole of oxygen (relative atomic mass = 16.0), do we mean 16.0 g or 32.0 g? To avoid ambiguity we need to specify whether we are talking about oxygen atoms or molecules.

Each oxygen molecule contains two oxygen atoms. So, one mole of oxygen molecules contains 2 moles of oxygen atoms. The mass of one mole of oxygen atoms is 16.0 g, so the mass of one mole of oxygen molecules is 32.0 g (i.e. 2 x 16.0 g).

Molar mass of a compound

The molar mass of a covalent compound is the mass of one mole of the compound. It is equal to the relative molecular mass of the compound expressed in grams per mole ($g\ mol^{-1}$).

The molar mass of an ionic compound is the mass of one mole of the compound. It is equal to the formula mass of the compound expressed in grams per mole ($g\ mol^{-1}$).

We can find the molar mass of a substance by the following steps:

1 Write down the symbol or chemical formula of the substance.

2 Find out the relative atomic mass, relative molecular mass or formula mass of the substance.

3 Add the units g mol^{-1} to the numerical value obtained.

Example 12.4

Q Calculate the molar masses of the following compounds:

a) sulphur dioxide (SO_2);

b) sucrose ($C_{12}H_{22}O_{11}$);

c) iron(III) sulphate ($Fe_2(SO_4)_3$).

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, S = 32.1, Fe = 55.8)

A a) Relative molecular mass of sulphur dioxide
= relative atomic mass of S + 2 x relative atomic mass of O
= 32.1 + 2 x 16.0
= 64.1

Molar mass of sulphur dioxide
= 64.1 g mol^{-1}

b) Relative molecular mass of sucrose
= 12 x relative atomic mass of C + 22 x relative atomic mass of H + 11 x relative atomic mass of O
= 12 x 12.0 + 22 x 1.0 + 11 x 16.0
= 342.0

Molar mass of sucrose = 342.0 g mol^{-1}

c) Formula mass of iron(III) sulphate
= 2 x relative atomic mass of Fe + 3 x (relative atomic mass of S + 4 x relative atomic mass of O)
= 2 x 55.8 + 3 x (32.1 + 4 x 16.0)
= 399.9

Molar mass of iron(III) sulphate
= 399.9 g mol^{-1}

Practice 12.2

Calculate the molar masses of the following substances:

a) nitrogen (N_2);
(2 x 14.0) g mol^{-1} = 28.0 g mol^{-1}

b) copper(II) nitrate ($Cu(NO_3)_2$);
[63.5 + 2 x (14.0 + 3 x 16.0)] g mol^{-1} = 187.5 g mol^{-1}

c) ammonium phosphate ($(NH_4)_3PO_4$);
[3 x (14.0 + 4 x 1.0) + 31.0 + 4 x 16.0] = 149.0 g mol^{-1}

d) sulphuric acid (H_2SO_4).
(2 x 1.0 + 32.1 + 4 x 16.0) g mol^{-1} = 98.1 g mol^{-1}

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, P = 31.0, S = 32.1, Cu = 63.5)

12.5 Calculations involving moles and masses

Converting mass to moles and moles to mass

We can work out the number of moles of a substance from its mass.

✓ Number of moles of a substance $= \dfrac{\text{mass of a substance (g)}}{\text{molar mass of the substance (g mol}^{-1}\text{)}}$

Example 12.5

Q What is the mass of 3.00 moles of chlorine molecules?

(Relative atomic masses: Cl = 35.5)

A Molar mass of chlorine molecules
= 2 x 35.5 g mol^{-1}
= 71.0 g mol^{-1}

Mass of chlorine molecules
= number of moles of chlorine molecules x molar mass of chlorine molecules
= 3.00 mol x 71.0 g mol^{-1}
= 213 g

∴ the mass of 3.00 moles of chlorine molecules is 213 g.

Example 12.6

Q What is the number of moles of potassium carbonate (K_2CO_3) present in 62.2 g of the substance?

(Relative atomic masses: C = 12.0, O = 16.0, K = 39.1)

A Molar mass of K_2CO_3
= (2 x 39.1 + 12.0 + 3 x 16.0) g mol^{-1}
= 138.2 g mol^{-1}

Number of moles of K_2CO_3

$$= \frac{\text{mass of } K_2CO_3}{\text{molar mass of } K_2CO_3}$$

$$= \frac{62.2 \text{ g}}{138.2 \text{ g mol}^{-1}}$$

= 0.450 mol

∴ there is 0.450 mole of potassium carbonate in 62.2 g of the substance.

Practice 12.3

1 What is the mass of 8.00 moles of methane (CH_4) molecules?

(Relative atomic masses: H = 1.0, C = 12.0)

2 What is the number of moles of silver oxide (Ag_2O) present in 81.2 g of the substance?

(Relative atomic masses: O = 16.0, Ag = 107.9)

Practice 12.3

1 Molar mass of CH_4 = (12.0 + 4 x 1.0) g mol^{-1}
= 16.0 g mol^{-1}

Mass of CH_4 = number of moles of CH_4 x molar mass of CH_4
= 8.00 mol x 16.0 g mol^{-1}
= 128 g

2 Molar mass of Ag_2O
= (2 x 107.9 + 16.0) g mol^{-1}
= 231.8 g mol^{-1}

$$\text{Number of moles of } Ag_2O = \frac{\text{mass of } Ag_2O}{\text{molar mass of } Ag_2O} = \frac{81.2 \text{ g}}{231.8 \text{ g mol}^{-1}} = 0.350 \text{ mol}$$

Calculating the number of particles in a sample

Since the number of moles of a substance relates to both its mass and to the number of particles in that substance, we can calculate one from the other.

Fig. 12.6 summarizes the steps in calculations involving the number of particles in a substance, the number of moles of the substance and the mass of the substance.

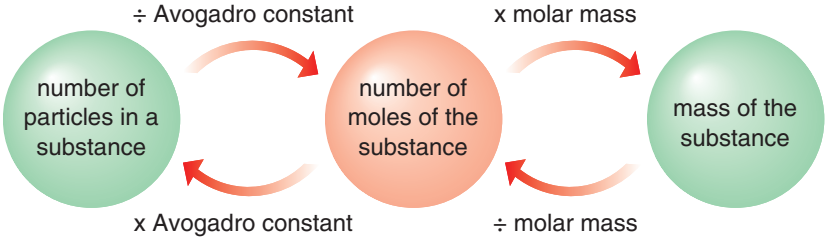


Fig. 12.6 The mole — linking quantities of a substance

Example 12.7

Q How many molecules are there in 81.6 g of ammonia (NH_3)? N1

(Relative atomic masses: H = 1.0, N = 14.0; Avogadro constant = 6.02×10^{23} mol^{-1})

A Molar mass of ammonia $= (14.0 + 3 \times 1.0)$ g mol^{-1}
$= 17.0$ g mol^{-1}

$$\text{Number of moles of ammonia} = \frac{\text{mass of } NH_3}{\text{molar mass of } NH_3}$$
$$= \frac{81.6 \text{ g}}{17.0 \text{ g mol}^{-1}}$$
$$= 4.80 \text{ mol}$$

Number of ammonia molecules = number of moles of ammonia x L
$= 4.80$ mol $\times 6.02 \times 10^{23}$ mol^{-1}
$= 2.89 \times 10^{24}$

$\therefore$ there are 2.89×10^{24} molecules in 81.6 g of ammonia.

N1 Examination questions may describe some samples and ask students to identify the sample that contains the greatest / smallest number of molecules (or atoms).

Example 12.8

Q A beaker contains 13.0 g of calcium phosphate ($Ca_3(PO_4)_2$). Calculate

a) the number of formula units of calcium phosphate present; and

b) the number of ions present.

(Relative atomic masses: O = 16.0, P = 31.0, Ca = 40.1; Avogadro constant = 6.02×10^{23} mol^{-1})

A a) Molar mass of $Ca_3(PO_4)_2$
$= [3 \times 40.1 + 2 \times (31.0 + 4 \times 16.0)]$ g mol^{-1}
$= 310.3$ g mol^{-1}

$$\text{Number of moles of } Ca_3(PO_4)_2 = \frac{\text{mass of } Ca_3(PO_4)_2}{\text{molar mass of } Ca_3(PO_4)_2}$$
$$= \frac{13.0 \text{ g}}{310.3 \text{ g mol}^{-1}}$$
$$= 0.0419 \text{ mol}$$

Number of formula units of $Ca_3(PO_4)_2$
= number of moles of $Ca_3(PO_4)_2$ x L
$= 0.0419$ mol $\times 6.02 \times 10^{23}$ mol^{-1}
$= 2.52 \times 10^{22}$

b) One formula unit of $Ca_3(PO_4)_2$ contains 5 ions. N2

Number of ions $= 5 \times 2.52 \times 10^{22}$
$= 1.26 \times 10^{23}$

$\therefore$ there are 2.52×10^{22} formula units of $Ca_3(PO_4)_2$ and 1.26×10^{23} ions in 13.0 g of calcium phosphate.

N2 One mole of $Ca_3(PO_4)_2$ contains three moles of calcium ions and two moles of phosphate ions.

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Example 12.9

Q The total volume of 20 drops of water is 1.00 cm^3. What is the number of molecules in 1 drop of water?

(Avogadro constant = 6.02 x 10^{23} mol^{-1}; density of water = 1.00 g cm^{-3}; relative atomic masses: H = 1.0, O = 16.0)

A Mass of 1.00 cm^3 of H_2O
= 1.00 g cm^{-3} x 1.00 cm^3
= 1.00 g

Mass of 1 drop of H_2O = $\frac{1.00\ g}{20}$
= 0.0500 g

Molar mass of H_2O = (2 x 1.0 + 16.0) g mol^{-1}
= 18.0 g mol^{-1}

Number of moles of molecules in 1 drop of water

$$= \frac{\text{mass of } H_2O}{\text{molar mass of } H_2O}$$

$$= \frac{0.0500\ g}{18.0\ g\ mol^{-1}}$$

= 2.78 x 10^{-3} mol

Number of molecules in 1 drop of water
= number of moles of H_2O x L
= 2.78 x 10^{-3} mol x 6.02 x 10^{23} mol^{-1}
= 1.67 x 10^{21}

∴ there are 1.67 x 10^{21} molecules in 1 drop of water.

Practice 12.4

Refer to page T34.

1 Consider 26.1 g of glucose ($C_6H_{12}O_6$).

a) What is the number of moles of glucose present?
0.145 mol

b) How many glucose molecules are present?
8.73 x 10^{22}

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0; Avogadro constant = 6.02 x 10^{23} mol^{-1})

2 Consider 84.7 g of zinc nitrate ($Zn(NO_3)_2$).

a) How many moles of zinc nitrate are present?
0.447 mol

b) How many zinc ions and nitrate ions are present respectively?
2.69 x 10^{23}, 5.38 x 10^{23}

(Relative atomic masses: N = 14.0, O = 16.0, Zn = 65.4; Avogadro constant = 6.02 x 10^{23} mol^{-1})



Fig. 12.7 A piece of haematite

12.6 Percentage by mass of an element in a compound

Fig. 12.7 shows a piece of haematite. The main iron compound in haematite is iron(III) oxide (Fe_2O_3).

Formula mass of Fe_2O_3
= 2 x relative atomic mass of Fe + 3 x relative atomic mass of O
= 2 x 55.8 + 3 x 16.0
= 159.6

One formula unit of Fe_2O_3 contains two iron(III) ions and three oxide ions. Iron contributes 111.6 units to the formula mass of Fe_2O_3.

Use formula mass when working with an ionic compound.
Use relative molecular mass when working with a covalent compound.

$$\text{Percentage by mass of iron in } Fe_2O_3 = \frac{111.6}{159.6} \times 100\%$$
$$= 69.9\%$$

We can work out the percentage by mass of each element in a compound using the following formula:

✓ Percentage by mass of element A in a compound

$$= \frac{\text{number of atoms of A in formula} \times \text{relative atomic mass of A}}{\text{formula mass or relative molecular mass of the compound}} \times 100\%$$

Example 12.10

Q Ammonium nitrate (NH_4NO_3) is used as a fertilizer due to the high usable nitrogen content. What is the percentage by mass of nitrogen in ammonium nitrate?

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0)

A Formula mass of NH_4NO_3 $= 2 \times 14.0 + 4 \times 1.0 + 3 \times 16.0$
$= 80.0$

Percentage by mass of N in NH_4NO_3

$$= \frac{\text{number of atoms of N in formula} \times \text{relative atomic mass of N}}{\text{formula mass of } NH_4NO_3} \times 100\%$$

$$= \frac{2 \times 14.0}{80.0} \times 100\%$$

$= 35.0\%$

$\therefore$ percentage by mass of nitrogen in ammonium nitrate is 35.0%.

Water of crystallization



Fig. 12.8 Blue hydrated copper(II) sulphate crystal

Some ionic compounds crystallize from a solution to form a **hydrated** ionic compound. In these compounds, water molecules are included in the crystal lattice.

For example, the formula of hydrated copper(II) sulphate is $CuSO_4 \bullet 5H_2O$ (Fig. 12.8). The crystal lattice is a regular pattern of three different particles: Cu^{2+} ions, SO_4^{2-} ions and H_2O molecules. The water is called **water of crystallization**.

We can calculate the percentage by mass of water in a hydrated compound from its formula.

Example 12.11

Q Calculate the percentage by mass of water in hydrated magnesium chloride ($MgCl_2 \bullet 6H_2O$).

(Relative atomic masses: H = 1.0, O = 16.0, Mg = 24.3, Cl = 35.5)

A Formula mass of $MgCl_2 \bullet 6H_2O = 24.3 + 2 \times 35.5 + 6 \times (2 \times 1.0 + 16.0)$
$= 203.3$

$$\text{Percentage by mass of water in } MgCl_2 \bullet 6H_2O = \frac{6 \times (2 \times 1.0 + 16.0)}{203.3} \times 100\%$$
$$= 53.1\%$$

$\therefore$ the percentage by mass of water in the hydrated magnesium chloride is 53.1%.

Example 12.12

Q Gypsum ($CaSO_4 \bullet nH_2O$) contains 20.9% by mass of water. What is the value of n?

(Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Ca = 40.1)

A Formula mass of $CaSO_4 \bullet nH_2O = 40.1 + 32.1 + 4 \times 16.0 + n(2 \times 1.0 + 16.0)$
$= 136.2 + 18n$

$$\text{Percentage by mass of water in } CaSO_4 \bullet nH_2O = \frac{18n}{136.2 + 18n} \times 100\%$$

$$\therefore \quad \frac{18n}{136.2 + 18n} \times 100\% = 20.9\%$$

$$n = 2$$

$\therefore$ gypsum contains two molecules of water of crystallization per formula unit.

hydrated 水合的　　water of crystallization 結晶水

Practice 12.5

Refer to page T35.

1 A lot of fertilizers contain ammonium sulphate ($(NH_4)_2SO_4$). What is the percentage by mass of nitrogen in the compound?
21.2%
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, S = 32.1)

2 What is the percentage by mass of oxygen in $CuSO_4 \bullet 5H_2O$?
57.7%
(Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Cu = 63.5)

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3 An ionic compound of chemical formula X_2O_3 contains 68.5% by mass of X. What is the relative atomic mass of X?
52.2
(Relative atomic mass: O = 16.0)

12.7 Determining the empirical formula of a compound

To determine the formula of a compound, chemists must perform an analysis to obtain information about the mass of each element present in a given mass of the compound. From the masses of the elements, chemists can calculate the mole ratio of atoms that combine to form the compound and determine the formula of the compound. The formula found is called the **empirical formula**.

Empirical means known by experiment.

✓ The empirical formula of a compound gives the simplest whole number ratio of atoms or ions present in the compound.

Fig. 12.9 shows the steps for determining the empirical formula of a compound.

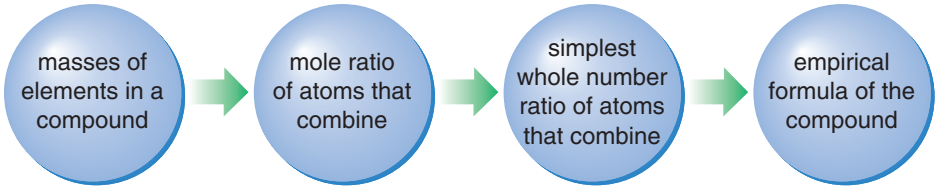


Fig. 12.9 Steps for determining the empirical formula of a compound

empirical formula 實驗式

12.1
Determining the empirical formula of magnesium oxide.

Determining the empirical formula of magnesium oxide

We can use the set-up shown in Fig. 12.10a to find the empirical formula of magnesium oxide.

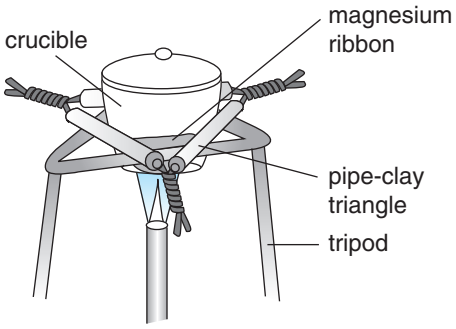


Fig. 12.10a Experimental set-up to find the empirical formula of magnesium oxide



Fig. 12.10b Magnesium oxide obtained after heating magnesium in a crucible

The following sample results are obtained.

Initial reading		Final reading	
Item	**Mass (g)**	**Item**	**Mass (g)**
Crucible + lid + magnesium	21.558	Crucible + lid + magnesium oxide	22.166
Crucible + lid	20.635	Magnesium oxide	22.166 – 20.635 = 1.531
Magnesium	21.558 – 20.635 = 0.923	Oxygen present in the oxide	1.531 – 0.923 = 0.608

	Magnesium	**Oxygen**
Mass of element in the compound	0.923 g	0.608 g
Relative atomic mass	24.3	16.0
Number of moles of atoms that combine	$\frac{0.923\ \text{g}}{24.3\ \text{g mol}^{-1}} = 0.0380\ \text{mol}$	$\frac{0.608\ \text{g}}{16.0\ \text{g mol}^{-1}} = 0.0380\ \text{mol}$
Mole ratio of atoms (obtained by dividing all numbers of moles by the smallest number of moles)	$\frac{0.0380\ \text{mol}}{0.0380\ \text{mol}} = 1$	$\frac{0.0380\ \text{mol}}{0.0380\ \text{mol}} = 1$

From the above results, we find that one mole of magnesium atoms combines with one mole of oxygen atoms, or one magnesium atom combines with one oxygen atom to give magnesium oxide. Hence the empirical formula of magnesium oxide is MgO.

Determining the empirical formula of red copper oxide

Fig. 12.11 shows the experimental set-up used to find the empirical formula of red copper oxide. Heat the weighed mass of the oxide in a combustion tube. Then pass hydrogen gas over it. The hydrogen gas reduces the red copper oxide to reddish brown copper.

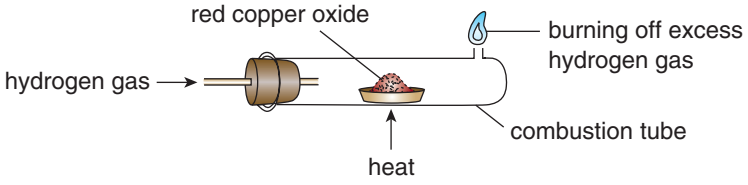


Fig. 12.11 Experimental set-up used to find the empirical formula of red copper oxide

The following sample results are obtained.

Initial reading		Final reading	
Item	**Mass (g)**	**Item**	**Mass (g)**
Combustion tube + red copper oxide	20.093	Combustion tube + copper	19.973
Combustion tube	19.020	Copper	19.973 – 19.020 = 0.953
Red copper oxide	20.093 – 19.020 = 1.073	Oxygen present in the oxide	1.073 – 0.953 = 0.120

	Copper	**Oxygen**
Mass of element in the compound	0.953 g	0.120 g
Relative atomic mass	63.5	16.0
Number of moles of atoms that combine	$\frac{0.953\ \text{g}}{63.5\ \text{g mol}^{-1}} = 0.0150\ \text{mol}$	$\frac{0.120\ \text{g}}{16.0\ \text{g mol}^{-1}} = 0.00750\ \text{mol}$
Mole ratio of atoms	$\frac{0.0150\ \text{mol}}{0.00750\ \text{mol}} = 2$	$\frac{0.00750\ \text{mol}}{0.00750\ \text{mol}} = 1$

From the above results, we find that the empirical formula of red copper oxide is Cu_2O.

Example 12.13

Q An oxide of metal M reacts completely with carbon to give 7.37 g of metal M and 4.36 g of carbon dioxide. What is the empirical formula of the oxide? N3

(Relative atomic masses: C = 12.0, O = 16.0, M = 55.8)

N3 Examination questions may give the actual amounts of X and oxygen that combine instead of the percentage by mass.

A Mass of oxygen in the oxide $= 4.36\text{ g} \times \frac{32.0}{44.0}$

$= 3.17\text{ g}$

	Metal M	Oxygen
Mass of element in the compound	7.37 g	3.17 g
Relative atomic mass	55.8	16.0
Number of moles of atoms that combine	$\frac{7.37\text{ g}}{55.8\text{ g mol}^{-1}} = 0.132\text{ mol}$	$\frac{3.17\text{ g}}{16.0\text{ g mol}^{-1}} = 0.198\text{ mol}$
Mole ratio of atoms	$\frac{0.132\text{ mol}}{0.132\text{ mol}} = 1.00$	$\frac{0.198\text{ mol}}{0.132\text{ mol}} = 1.50$
Simplest whole number ratio of atoms	1.00 x 2 = 2	1.50 x 2 = 3

∴ the empirical formula of the oxide is M_2O_3.

Working out an empirical formula using percentage composition

N4 Examination questions may give the percentage composition of a compound and ask students to calculate the mole ratio of elements in the compound.

Often, information for the composition of compounds is given as percentages by mass. For example, in a sulphide of titanium, the mass percentage of titanium is 42.7%. Thus, the mass of percentage of sulphur is 57.3% (i.e. 100% – 42.7%). N4

These percentage figures apply to any amount of substance we choose — so choose 100.0 g. In which case, the percentages convert simply into masses.

CE 2008 Paper 2 Q31
AL 2007 Paper 2 Q7(c)(i)

	Titanium	Sulphur
Mass of element in the compound	42.7 g	57.3 g
Relative atomic mass	47.9	32.1
Number of moles of atoms that combine	$\frac{42.7\text{ g}}{47.9\text{ g mol}^{-1}} = 0.891\text{ mol}$	$\frac{57.3\text{ g}}{32.1\text{ g mol}^{-1}} = 1.79\text{ mol}$
Mole ratio of atoms	$\frac{0.891\text{ mol}}{0.891\text{ mol}} = 1$	$\frac{1.79\text{ mol}}{0.891\text{ mol}} = 2$

Thus, the empirical formula of the sulphide is TiS_2.

Practice 12.6

Refer to page T36.

1 Complete reduction of 41.1 g of an oxide of metal X by hydrogen gas produces metal X and 4.28 g of water. What is the empirical formula of the oxide? N5

(Relative atomic masses: H = 1.0, O = 16.0, X = 207.2)

2 An oxide of manganese contains 63.2% of manganese by mass.

Calculate the empirical formula of the oxide.

(Relative atomic masses: O = 16.0, Mn = 54.9)

N5 Examination questions may give the mass of water obtained upon reduction of a metal oxide by hydrogen and ask students to calculate the relative atomic mass of the metal.

CE 2011 Paper 2 Q38
DSE 2012 Paper 1A Q3

3 Hydrogen is passed over heated oxide of metal M to reduce the oxide to metal M. The following results are obtained:

Mass of empty combustion tube = 52.2 g

Mass of combustion tube + oxide (before experiment)
= 68.1 g

Mass of combustion tube + metal M (after experiment)
= 64.9 g

What is the empirical formula of the oxide of metal M?

(Relative atomic masses: O = 16.0, M = 63.5)

Molecular formula and empirical formula

The empirical formula of a covalent compound shows the simplest whole number ratio of atoms in the compound. This contrasts with the molecular formula, which gives the actual number of atoms in a molecule. Table 12.3 lists the molecular and empirical formulae of some covalent compounds. For many covalent compounds, such as carbon dioxide and water, the molecular and the empirical formulae are the same.

Table 12.1

Molecular and empirical formulae of some covalent compounds

Covalent compound	Molecular formula	Empirical formula
Glucose	$C_6H_{12}O_6$	CH_2O
Vitamin C	$C_6H_8O_6$	$C_3H_4O_3$
Carbon dioxide	CO_2	CO_2
Water	H_2O	H_2O

Example 12.14

Q A compound X contains 85.7% carbon and 14.3% hydrogen. Its relative molecular mass is 56.0. Find

a) its empirical formula; and

b) its molecular formula. ASL 2005 Q6(a)

(Relative atomic masses: H = 1.0, C = 12.0)

CE 2008 Paper 2 Q31
CE 2011 Paper 2 Q38

A a) Suppose we have 100.0 g of compound X, containing 85.7 g of carbon and 14.3 g of hydrogen.

	Carbon	Hydrogen
Mass of element in the compound	85.7 g	14.3 g
Relative atomic mass	12.0	1.0
Number of moles of atoms that combine	$\frac{85.7\text{ g}}{12.0\text{ g mol}^{-1}} = 7.14\text{ mol}$	$\frac{14.3\text{ g}}{1.0\text{ g mol}^{-1}} = 14.3\text{ mol}$
Mole ratio of atoms	$\frac{7.14\text{ mol}}{7.14\text{ mol}} = 1$	$\frac{14.3\text{ mol}}{7.14\text{ mol}} = 2$

$\therefore$ the empirical formula of compound X is CH_2.

Continued on next page

glucose 葡萄糖

b) Let $(CH_2)_n$ be the molecular formula of compound X.

Relative molecular mass of X = $n(12.0 + 2 \times 1.0)$
= $14n$
$\therefore$ $14n = 56.0$
$n = 4$

$\therefore$ the molecular formula of compound X is $(CH_2)_4$ or C_4H_8.

Working out the formula of a hydrated salt

When hydrated crystals are heated, the water is driven off as steam, leaving behind an anhydrous solid. For example, blue hydrated copper(II) sulphate crystals ($CuSO_4 \bullet 5H_2O$) form white **anhydrous** copper(II) sulphate ($CuSO_4$) on heating (Fig. 12.12).



Fig. 12.12a Blue hydrated copper(II) sulphate crystals



Fig. 12.12b When hydrated copper(II) sulphate crystals are heated, the water is driven off

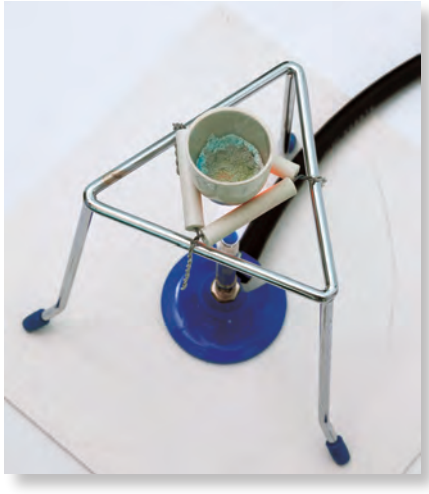


Fig. 12.12c White anhydrous copper(II) sulphate is formed

anhydrous 無水的

Example 12.15

Q A sample of hydrated cobalt chloride ($CoCl_2 \bullet nH_2O$) is heated in a crucible to drive off the water. The following results are obtained:

Mass of crucible + lid = 20.52 g

Mass of crucible + lid + hydrated salt = 56.20 g

Mass of crucible + lid + anhydrous salt = 40.01 g

Work out the formula of the hydrated cobalt chloride.

(Relative atomic masses: H = 1.0, O = 16.0, Cl = 35.5, Co = 58.9)

A Mass of anhydrous salt = (40.01 – 20.52) g = 19.49 g

Mass of water = (56.20 – 40.01) g = 16.19 g

	$CoCl_2$	H_2O
Mass of compound in the hydrated salt	19.49 g	16.19 g
Formula mass / relative molecular mass	58.9 + 2 x 35.5 = 129.9	2 x 1.0 + 16.0 = 18.0
Number of moles of compound in the hydrated salt	$\frac{19.49\text{ g}}{129.9\text{ g mol}^{-1}} = 0.150\text{ mol}$	$\frac{16.19\text{ g}}{18.0\text{ g mol}^{-1}} = 0.899\text{ mol}$
Mole ratio of compounds	$\frac{0.150\text{ mol}}{0.150\text{ mol}} = 1$	$\frac{0.899\text{ mol}}{0.150\text{ mol}} = 6$

∴ the formula of the hydrated salt is $CoCl_2 \bullet 6H_2O$.

Practice 12.7

Refer to page T37.

1 A chemical analysis of caffeine indicates its composition to be 49.5% carbon, 5.2% hydrogen, 16.5% oxygen and 28.9% nitrogen by mass.

a) Calculate the empirical formula of caffeine.

b) The relative molecular mass of caffeine is 194.0. What is its molecular formula?

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0)

2 The formula for hydrated sodium carbonate is $Na_2CO_3 \bullet xH_2O$. On strong heating, 14.3 g of the carbonate produce 9.00 g of water. What is the value of x?

10

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0)

12.8 Mole ratio in a balanced chemical equation

We can use the following balanced chemical equation to represent the reaction between hydrogen and oxygen:

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$$

From the equation, we know that 2 hydrogen molecules react with 1 oxygen molecule to give 2 water molecules. Thus, $2 \times 6.02 \times 10^{23}$ hydrogen molecules react with 6.02×10^{23} oxygen molecules to give $2 \times 6.02 \times 10^{23}$ water molecules. That means 2 moles of hydrogen molecules react with 1 mole of oxygen molecules to give 2 moles of water molecules.

$2H_2(g)$	+	$O_2(g)$	$\longrightarrow$	$2H_2O(l)$
2 hydrogen molecules		1 oxygen molecule		2 water molecules
$2 \times 6.02 \times 10^{23}$ hydrogen molecules		6.02×10^{23} oxygen molecules		$2 \times 6.02 \times 10^{23}$ water molecules
2 moles of hydrogen molecules		1 mole of oxygen molecules		2 moles of water molecules

Thus, the coefficients before the reactant(s) and product(s) in a chemical equation indicate the mole ratio of the reactant(s) and product(s).

12.9 Chemical equations and reacting masses

A chemical equation indicates the relationship between the number of moles of reactant(s) and product(s) in a chemical reaction. Since the masses of substances are related to the numbers of moles, we can use a chemical equation to determine the relationship between the masses of reactant(s) and product(s).

Consider the reaction between hydrogen and oxygen:

$2H_2(g)$	+	$O_2(g)$	$\longrightarrow$	$2H_2O(l)$
2 moles		1 mole		2 moles

The molar masses of H_2, O_2 and H_2O are 2.0 g mol^{-1}, 32.0 g mol^{-1} and 18.0 g mol^{-1} respectively.

The equation tells us that:

4.0 g of hydrogen	react with	32.0 g of oxygen	to give	36.0 g of water

We can also write

2.0 g of hydrogen	react with	16.0 g of oxygen	to give	18.0 g of water

Mass-mass calculations from chemical equations

Follow the steps shown below when doing calculations from equations (Fig. 12.13).

1 Write down the balanced chemical equation for the reaction.

$aA + bB \longrightarrow cC + dD$

2 Write down the amounts of substances given in the question underneath their formulae.

3 Suppose that we are given the mass of A. Then convert this to the number of moles.

$$\text{Number of moles of A} = \frac{\text{mass of A (g)}}{\text{molar mass of A (g mol}^{-1}\text{)}}$$

4 Use the balanced chemical equation to work out the number of moles of the substance required in the question.

The equation tells us that '*a*' moles of A react with '*b*' moles of B to form '*c*' moles of C and '*d*' moles of D.

5 Convert the number of moles of the substance required to mass.

$$\text{Mass of the substance (g)} = \text{number of moles of the substance (mol)} \times \text{molar mass of the substance (g mol}^{-1}\text{)}$$ N6

N6 Reaction between equal masses of Mg and Zn with dilute acid

Suppose equal masses of different metals, such as Mg and Zn, are added separately to excess dilute H_2SO_4 / HCl. One of the metals (Mg in this case) produces a greater amount of hydrogen. Examination questions may ask what can be deduced from that (CE 2007 Paper 2 Q5 and CE 2008 Paper 2 Q50).

This may be explained using a concrete example. Suppose 100 g of Mg and 100 g of Zn are added separately to excess dilute H_2SO_4.

$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(g)$

$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$

$$\text{Number of moles of Mg in 100 g} = \frac{\text{mass of Mg}}{\text{molar mass of Mg}} = \frac{100\text{ g}}{24.3\text{ g mol}^{-1}} = 4.12\text{ mol}$$

$$\text{Number of moles of Zn in 100 g} = \frac{\text{mass of Zn}}{\text{molar mass of Zn}} = \frac{100\text{ g}}{65.4\text{ g mol}^{-1}} = 1.53\text{ mol}$$

100 g of Mg will produce 4.12 moles of hydrogen while 100 g of Zn will produce 1.53 moles of hydrogen. Hence the difference in the amount of hydrogen produced by Mg and Zn is due to the difference between the molar masses of the two metals.

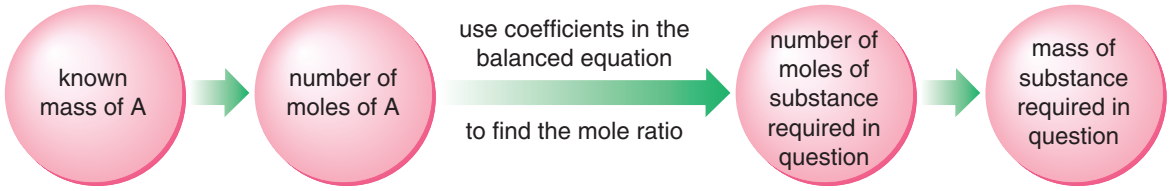


Fig. 12.13 Steps to work out the reacting masses of substances

Example 12.16

Q Copper(I) sulphide reacts with oxygen according to the following chemical equation:

$$Cu_2S(s) + O_2(g) \longrightarrow 2Cu(s) + SO_2(g)$$

Calculate the mass of copper formed when 82.7 g of copper(I) sulphide react completely with oxygen.

(Relative atomic masses: S = 32.1, Cu = 63.5)

CE 2006 Paper 1 Q5(d)

A Method 1

1 $Cu_2S(s) + O_2(g) \longrightarrow 2Cu(s) + SO_2(g)$

2 82.7 g ? g

3 Molar mass of Cu_2S = (2 x 63.5 + 32.1) g mol^{-1}
= 159.1 g mol^{-1}

$$\text{Number of moles of } Cu_2S = \frac{\text{mass of } Cu_2S}{\text{molar mass of } Cu_2S}$$

$$= \frac{82.7 \text{ g}}{159.1 \text{ g mol}^{-1}}$$

$$= 0.520 \text{ mol}$$

4 According to the equation, 1 mole of Cu_2S reacts with 1 mole of O_2 to give 2 moles of Cu.

$\therefore$ number of moles of Cu formed = 2 x 0.520 mol
= 1.04 mol

5 Molar mass of Cu = 63.5 g mol^{-1}

Mass of Cu formed = number of moles of Cu x molar mass of Cu
= 1.04 mol x 63.5 g mol^{-1}
= 66.0 g

Continued on next page

Method 2

$$\underset{82.7\text{ g}}{Cu_2S(s)} + O_2(g) \longrightarrow \underset{?\text{ g}}{2Cu(s)} + SO_2(g)$$

Molar mass of $Cu_2S = (2 \times 63.5 + 32.1)\text{ g mol}^{-1}$
$= 159.1\text{ g mol}^{-1}$

Molar mass of $Cu = 63.5\text{ g mol}^{-1}$

According to the equation, 1 mole of Cu_2S reacts with 1 mole of O_2 to give 2 moles of Cu.

$\therefore$ 159.1 g of Cu_2S react to give 2×63.5 g of Cu.

$$\underset{\substack{159.1\text{ g}\\82.7\text{ g}}}{Cu_2S(s)} + O_2(g) \longrightarrow \underset{\substack{2 \times 63.5\text{ g}\\?\text{ g}}}{2Cu(s)} + SO_2(g)$$

$$\text{Mass of Cu formed} = 82.7\text{ g} \times \frac{2 \times 63.5\text{ g}}{159.1\text{ g}}$$
$$= 66.0\text{ g}$$

$\therefore$ the mass of copper formed was 66.0 g.

Example 12.17

Q Traditional gunpowder is a mixture of potassium nitrate, sulphur powder and charcoal powder. When this gunpowder is ignited, solid potassium sulphide, nitrogen gas and carbon dioxide gas are formed.

a) Complete and balance the chemical equation for the reaction that occurs upon the ignition of the gunpowder.

_____ $KNO_3(s)$ + $S(s)$ + _____ $C(s) \longrightarrow$

b) What is the mass of charcoal required to react with 75.5 g of potassium nitrate? CE 2005 Paper 2 Q37

(Relative atomic masses: C = 12.0, N = 14.0, O = 16.0, K = 39.1)

A a) $2KNO_3(s) + S(s) + 3C(s) \longrightarrow K_2S(s) + N_2(g) + 3CO_2(g)$

b) Method 1

1 $2KNO_3(s) + S(s) + 3C(s) \longrightarrow K_2S(s) + N_2(g) + 3CO_2(g)$
2 75.5 g ? g

Continued on next page

3 Molar mass of KNO_3 = (39.1 + 14.0 + 3 x 16.0) g mol^{-1}
= 101.1 g mol^{-1}

$$\text{Number of moles of } KNO_3 = \frac{\text{mass of } KNO_3}{\text{molar mass of } KNO_3}$$
$$= \frac{75.5 \text{ g}}{101.1 \text{ g mol}^{-1}}$$
$$= 0.747 \text{ mol}$$

4 According to the equation, 2 moles of KNO_3 require 3 moles of C for reaction.

$$\therefore \text{ number of moles of C required} = \frac{3}{2} \times 0.747 \text{ mol}$$
$$= 1.12 \text{ mol}$$

5 Molar mass of C = 12.0 g mol^{-1}

Mass of C required = number of moles of C x molar mass of C
= 1.12 mol x 12.0 g mol^{-1}
= 13.4 g

Method 2

Molar mass of KNO_3 = (39.1 + 14.0 + 3 x 16.0) g mol^{-1}
= 101.1 g mol^{-1}

Molar mass of C = 12.0 g mol^{-1}

According to the equation, 2 moles of KNO_3 require 3 moles of C for reaction.

$\therefore$ 2 x 101.1 g of KNO_3 require 3 x 12.0 g of C for reaction.

$$\underset{\substack{2 \times 101.1 \text{ g} \\ 75.5 \text{ g}}}{2KNO_3(s)} + S(s) + \underset{\substack{3 \times 12.0 \text{ g} \\ ? \text{ g}}}{3C(s)} \longrightarrow K_2S(s) + N_2(g) + 3CO_2(g)$$

$$\text{Mass of C required} = 75.5 \text{ g} \times \frac{3 \times 12.0 \text{ g}}{2 \times 101.1 \text{ g}}$$
$$= 13.4 \text{ g}$$

$\therefore$ 13.4 g of charcoal are required.

Example 12.18

Q Methane (CH_4) is one of the components of the gaseous fuel used in a Bunsen burner. When methane burns completely, carbon dioxide and water are formed.

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$

What is the mass of methane that can give 1.35 g of water?

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

CE 2007 Paper 2 Q34
CE 2010 Paper 2 Q14

A Method 1

1 $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$
2 ? g 1.35 g

3 Molar mass of $H_2O = (2 \times 1.0 + 16.0)$ g mol^{-1}
$= 18.0$ g mol^{-1}

$$\text{Number of moles of } H_2O = \frac{\text{mass of } H_2O}{\text{molar mass of } H_2O}$$
$$= \frac{1.35 \text{ g}}{18.0 \text{ g mol}^{-1}}$$
$$= 0.0750 \text{ mol}$$

4 According to the equation, 1 mole of CH_4 gives 2 moles of H_2O.

$$\therefore \text{ number of moles of } CH_4 = \frac{0.0750}{2} \text{ mol}$$
$$= 0.0375 \text{ mol}$$

5 Molar mass of $CH_4 = (12.0 + 4 \times 1.0)$ g mol^{-1}
$= 16.0$ g mol^{-1}

Mass of CH_4 = number of moles of CH_4 x molar mass of CH_4
$= 0.0375$ mol x 16.0 g mol^{-1}
$= 0.600$ g

Continued on next page

Method 2

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$
? g 1.35 g

Molar mass of $CH_4 = (12.0 + 4 \times 1.0)\ g\ mol^{-1}$
$= 16.0\ g\ mol^{-1}$

Molar mass of $H_2O = (2 \times 1.0 + 16.0)\ g\ mol^{-1}$
$= 18.0\ g\ mol^{-1}$

According to the equation, 1 mole of CH_4 gives 2 moles of H_2O.

$\therefore$ 16.0 g of CH_4 give 2×18.0 g of H_2O.

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$
16.0 g 2 x 18.0 g
? g 1.35 g

$$\text{Mass of } CH_4 = 1.35\ g \times \frac{16.0\ g}{2 \times 18.0\ g}$$
$$= 0.600\ g$$

$\therefore$ 0.600 g of methane can give 1.35 g of water.

Practice 12.8

1 Sodium azide (NaN_3) is used in airbags in cars. When there is a serious collision, the azide will decompose to give nitrogen. The decomposition can be represented by the equation:

$$2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$$

What is the mass of nitrogen produced when 7.80 g of sodium azide decompose?
5.04 g; refer to page T38.
(Relative atomic masses: N = 14.0, Na = 23.0)

2 Thermit reactions broadly refer to reactions between a metal powder and a metal oxide. One example is the reaction of iron(III) oxide with aluminium.

a) Complete and balance the chemical equation for the following thermit reaction.

1 $Fe_2O_3(s)$ + 2 $Al(s) \longrightarrow$ $2Fe(s) + Al_2O_3(s)$

b) Calculate the mass of aluminium required to react completely with 63.8 g of iron(III) oxide.
21.6 g; refer to page T38.
(Relative atomic masses: O = 16.0, Al = 27.0, Fe = 55.8)

DSE 2013 Paper 1B Q7(a)(i)

Example 12.19

Q In an oxide of metal M, the ion of M carries +1 charge. Complete decomposition of 3.48 g of the oxide gives 3.24 g of metal M. What is the relative atomic mass of M?

(Relative atomic masses: O = 16.0)

CE 2010 Paper 1 Q4(b)

A Let x be the relative atomic mass of M.

The chemical formula of the oxide of M is M_2O.

$$\underset{3.48\ g}{2M_2O(s)} \longrightarrow \underset{3.24\ g}{4M(s)} + O_2(g)$$

Molar mass of $M_2O = (2x + 16.0)\ g\ mol^{-1}$

$$\text{Number of moles of } M_2O = \frac{3.48\ g}{(2x + 16.0)\ g\ mol^{-1}}$$

$$\text{Number of moles of M} = \frac{3.24\ g}{x\ g\ mol^{-1}}$$

According to the equation, 2 moles of M_2O decompose to give 4 moles of M.

i.e. 2 x number of moles of M_2O = number of moles of M

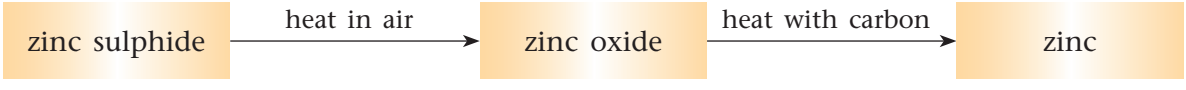
$$2 \times \frac{3.48}{(2x + 16.0)} = \frac{3.24}{x}$$

$$x = 108$$

$\therefore$ the relative atomic mass of M is 108.

Example 12.20

Q The flow diagram below shows the stages involved in the extraction of zinc from an ore of zinc sulphide.



An ore contains 75.0% of zinc sulphide by mass. Assuming that the other components in this ore do not contain zinc, what mass of the ore is required to extract 15.7 g of zinc?

(Relative atomic masses: S = 32.1, Zn = 65.4)

Continued on next page

CE 2011 Paper 2 Q8

A 1 $ZnS(s) \longrightarrow ZnO(s) \longrightarrow Zn(s)$
2 ? g 15.7 g

3 Molar mass of Zn = 65.4 g mol^{-1}

$$\text{Number of moles of Zn} = \frac{\text{mass of Zn}}{\text{molar mass of Zn}}$$
$$= \frac{15.7 \text{ g}}{65.4 \text{ g mol}^{-1}}$$
$$= 0.240 \text{ mol}$$

4 1 mole of Zn can be extracted from 1 mole of ZnS.

∴ number of moles of ZnS required = 0.240 mol

5 Molar mass of ZnS = (65.4 + 32.1) g mol^{-1}
= 97.5 g mol^{-1}

Mass of ZnS required = number of moles of ZnS x molar mass of ZnS
= 0.240 mol x 97.5 g mol^{-1}
= 23.4 g

$$\text{Mass of ore required} = \frac{23.4 \text{ g}}{0.750}$$
$$= 31.2 \text{ g}$$

∴ 31.2 g of the ore are required.

Practice 12.9

1 An oxide of M (MO) reacts completely with carbon to give 12.4 g of metal M and 4.29 g of carbon dioxide.

a) Write a chemical equation for the reaction between the oxide of M and carbon.
$2MO(s) + C(s) \longrightarrow 2M(s) + CO_2(g)$

b) What is the relative atomic mass of M?
63.6; refer to page T39.

(Relative atomic masses: C = 12.0, O = 16.0)

2 A sample of 3.80 g of impure calcium carbonate was allowed to react with excess dilute hydrochloric acid. It was found that 1.43 g of carbon dioxide were given off in the reaction.
Refer to page T40.

a) Write a word equation for the reaction of calcium carbonate with dilute hydrochloric acid.

b) Transcribe the word equation in (a) into a chemical equation.

c) Calculate the percentage by mass of calcium carbonate in the sample.
85.5%

(Relative atomic masses: C = 12.0, O = 16.0, Ca = 40.1)

Animation
Limiting reactants

12.10 Limiting reactant N7

In a chemical reaction, the reactant that is totally consumed when the reaction completes is called the **limiting reactant**. The amount of product formed is limited by this reactant since the reaction cannot proceed further without it. The other reactants in the reaction are in excess. Often, in a chemical reaction, it is preferred that the most valuable reactant is all consumed when the reaction completes. In order to do that, chemists add other reactants in the reaction in excess to achieve the maximum yield when the reaction completes.

N7 Suppose N moles of $Fe_2O_3(s)$ and 2N moles of CO(g) are allowed to react.

$Fe_2O_3(s) + 3CO(g) \longrightarrow 2Fe(l) + 3CO_2(g)$
N mol 2N mol

According to the equation, 1 mole of Fe_2O_3 reacts with 3 moles of CO to produce 2 moles of Fe. During the reaction, 2N moles of CO react with $\frac{2}{3}$N mole of Fe_2O_3. Thus, Fe_2O_3 was in excess. CO was the limiting reactant.

Number of moles of Fe produced
$= \frac{2}{3} \times 2N$ mol
$= \frac{4}{3}N$ mol

Example 12.21

Q Titanium is prepared by the reaction of titanium(IV) chloride with molten magnesium between 950 °C and 1 150 °C. The reaction can be represented by the equation:

$$TiCl_4(g) + 2Mg(l) \longrightarrow Ti(s) + 2MgCl_2(l)$$

What mass of titanium can be produced by heating 64.6 g of titanium(IV) chloride with 19.2 g of magnesium? N8

(Relative atomic masses: Mg = 24.3, Cl = 35.5, Ti = 47.9)

N8 Examination questions may NOT tell students when a limiting reactant is involved. Be careful when the question gives the quantities of both reagents.

A

1 $TiCl_4(g) + 2Mg(l) \longrightarrow Ti(s) + 2MgCl_2(l)$
2 64.6 g 19.2 g ? g

3 Molar mass of $TiCl_4 = (47.9 + 4 \times 35.5)$ g mol^{-1}
$= 189.9$ g mol^{-1}

$$\text{Number of moles of } TiCl_4 \text{ present} = \frac{\text{mass of } TiCl_4}{\text{molar mass of } TiCl_4}$$
$$= \frac{64.6 \text{ g}}{189.9 \text{ g mol}^{-1}}$$
$$= 0.340 \text{ mol}$$

$$\text{Number of moles of Mg present} = \frac{\text{mass of Mg}}{\text{molar mass of Mg}}$$
$$= \frac{19.2 \text{ g}}{24.3 \text{ g mol}^{-1}}$$
$$= 0.790 \text{ mol}$$

Continued on next page

limiting reactant 限量反應物

4 According to the equation, 1 mole of $TiCl_4$ reacts with 2 moles of Mg to produce 1 mole of Ti. During the reaction, 0.340 mole of $TiCl_4$ reacted with 0.680 mole of Mg. Thus, Mg was in excess. The amount of $TiCl_4$ limited the amount of Ti produced.

Number of moles of Ti produced = 0.340 mol

5 Mass of Ti produced = number of moles of Ti x molar mass of Ti
= 0.340 mol x 47.9 g mol^{-1}
= 16.3 g

∴ 16.3 g of titanium are produced.

Practice 12.10

Refer to page T40.

When heated, lithium reacts with nitrogen to form lithium nitride as shown in the equation below:

$6Li(s) + N_2(g) \longrightarrow 2Li_3N(s)$

8.28 g of lithium were heated with 10.6 g of nitrogen. Calculate the mass of lithium nitride formed.
13.9 g
(Relative atomic masses: Li = 6.9, N = 14.0)

N9 Difference between theoretical yield and actual yield

Often when performing chemical reactions in the laboratory or in industry, less product is obtained than expected because:

- the product does not all precipitate out of the solution; or
- there are physical losses (such as little splashes, inefficient rinsing, inefficient filtering or leakage of gases); or
- some of the reactants or products are lost by vaporization; or
- not all the reactant actually reacts (because it is covered with a product); or
- the reaction becomes too slow for us to wait for it to go to completion; or
- there is a side reaction which uses up some of the reactants without producing the desired product.

12.11 Theoretical yield and percentage yield N9

Iron is extracted from iron ore in a special furnace called a blast furnace. A typical blast furnace can produce up to 10 000 tonnes of molten iron a day. Suppose we want to find out how much iron can be produced from 10.0 tonnes of iron(III) oxide.

In a blast furnace, iron(III) oxide reacts with carbon monoxide according to the following equation:

$$\underset{\text{10.0 tonnes}}{Fe_2O_3(s)} + 3CO(g) \longrightarrow \underset{\text{? tonnes}}{2Fe(s)} + 3CO_2(g)$$

Molar mass of Fe_2O_3 = (2 x 55.8 + 3 x 16.0) g mol^{-1}
= 159.6 g mol^{-1}

Molar mass of Fe = 55.8 g mol^{-1}

According to the equation, 1 mole of Fe_2O_3 produces 2 moles of Fe.

∴ 159.6 g of Fe_2O_3 produce 2 x 55.8 g of iron.

Mass of iron produced from 10.0 tonnes of iron(III) oxide

$$= 10.0 \text{ tonnes} \times \frac{2 \times 55.8 \text{ g}}{159.6 \text{ g}}$$

$$= 6.99 \text{ tonnes}$$

We find that 10.0 tonnes of iron(III) oxide can give 6.99 tonnes of iron. The mass of iron produced is known as the **yield**, but we could never achieve this yield in a real blast furnace. Thus, we say that the calculation from the balanced equation gives the **theoretical yield**, which is the maximum amount possibly obtained if all the reactants are converted into products.

The **actual yield** from a reaction can be found only by carrying out the reaction. From 10.0 tonnes of iron(III) oxide, there will be less than 6.99 tonnes of iron produced. **Percentage yield** is a way of expressing the efficiency of a reaction.

There are many reasons why the actual yield of a reaction is not 100% in real situation. For example, not all the iron(III) oxide reacts. The iron(III) oxide may also contain impurities.

✓ $$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Example 12.22

Q Ammonia is produced from nitrogen and hydrogen according to the following chemical equation:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

a) Calculate the theoretical yield of ammonia produced when 7.80 tonnes of hydrogen react with excess nitrogen.

b) The percentage yield of ammonia is 15.0%. What is the actual yield of ammonia?

(Relative atomic masses: H = 1.0, N = 14.0)

Continued on next page

yield 產量　theoretical yield 理論產量　actual yield 實際產量　percentage yield 百分產率

A1 a) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

7.80 tonnes ? tonnes

Molar mass of H_2 = (2 x 1.0) g mol^{-1}
= 2.0 g mol^{-1}

Molar mass of NH_3 = (14.0 + 3 x 1.0) g mol^{-1}
= 17.0 g mol^{-1}

According to the equation, 3 moles of H_2 react with excess N_2 to produce 2 moles of NH_3.

∴ 3 x 2.0 g of H_2 react with excess N_2 to produce 2 x 17.0 g of NH_3.

$$\text{Theoretical yield of } NH_3 = 7.80 \text{ tonnes} \times \frac{2 \times 17.0 \text{ g}}{3 \times 2.0 \text{ g}}$$

= 44.2 tonnes

b) Actual yield of NH_3 = 44.2 tonnes x 15.0%
= 6.63 tonnes

∴ the theoretical yield and actual yield of ammonia are 44.2 tonnes and 6.63 tonnes respectively.

Practice 12.11

A lime kiln produces 500 kg of quicklime (calcium oxide) from 1 000 kg of limestone (calcium carbonate).

a) Write a chemical equation for the decomposition of limestone.
$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

b) What is the percentage yield of quicklime?

(Relative atomic masses: C = 12.0, O = 16.0, Ca = 40.1)

b) Molar mass of $CaCO_3$ = (40.1 + 12.0 + 3 x 16.0) g mol^{-1}
= 100.1 g mol^{-1}

Molar mass of CaO = (40.1 + 16.0) g mol^{-1}
= 56.1 g mol^{-1}

According to the equation, 1 mole of $CaCO_3$ produces 1 mole of CaO.

∴ 100.1 g of $CaCO_3$ decompose to produce 56.1 g of CaO.

$$\text{Theoretical yield of CaO} = 1\,000 \text{ kg} \times \frac{56.1 \text{ g}}{100.1 \text{ g}}$$

= 560 kg

$$\text{Percentage yield of CaO} = \frac{500 \text{ kg}}{560 \text{ kg}} \times 100\%$$

= 89.3%

∴ the percentage yield of quicklime is 89.3%.

Have you mastered?

Key terms

mole 摩爾 70 Avogadro constant 亞佛加德羅常數 70 molar mass 摩爾質量 73

hydrated 水合的 80 water of crystallization 結晶水 80 empirical formula 實驗式 81

anhydrous 無水的 87 limiting reactant 限量反應物 98 yield 產量 100

theoretical yield 理論產量 100 actual yield 實際產量 100 percentage yield 百分產率 100

Checklist

After studying this unit, you should be able to

- ☐ understand and use the quantitative information provided by a balanced chemical equation;
- ☐ perform calculations related to moles, Avogadro constant and molar masses;
- ☐ calculate the percentage by mass of an element in a compound using appropriate information;
- ☐ determine empirical formulae and molecular formulae from compositions by mass and molar masses;
- ☐ calculate masses of reactants and products in a reaction from the relevant equation and state the interrelationship between them;
- ☐ solve problems involving limiting reagents;
- ☐ calculate theoretical yield and percentage yield.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 One mole (symbol: mol) is the amount of a substance that contains the same number of particles as there are atoms in 12.0 g of carbon-12 ($^{12}_{6}C$).

2 Experiments show that 12.0 g of carbon-12 ($^{12}_{6}C$) contains 6.02×10^{23} atoms. Chemists call this number the Avogadro constant (symbol: L).

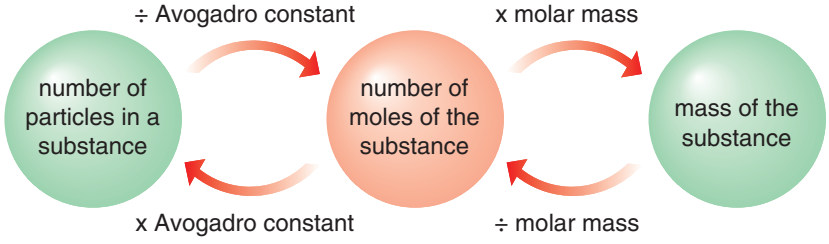
$$L = 6.02 \times 10^{23} \text{ mol}^{-1}$$

3 Number of particles in a substance = number of moles of the substance x L

4 The mass of one mole of a substance is called its molar mass. The units of molar mass are gram per mole ($g\ mol^{-1}$).

5 $$\text{Number of moles of a substance} = \frac{\text{mass of a substance (g)}}{\text{molar mass of the substance (g mol}^{-1}\text{)}}$$

6 The following diagram summarizes the steps in calculations involving the number of particles in a substance, the number of moles of the substance and the mass of the substance.

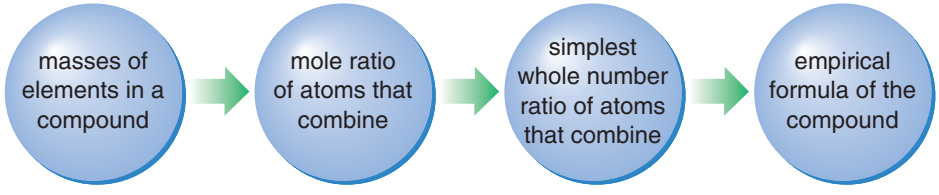


7 Percentage by mass of element A in a compound

$$= \frac{\text{number of atoms of A in formula x relative atomic mass of A}}{\text{formula mass or relative molecular mass of the compound}} \times 100\%$$

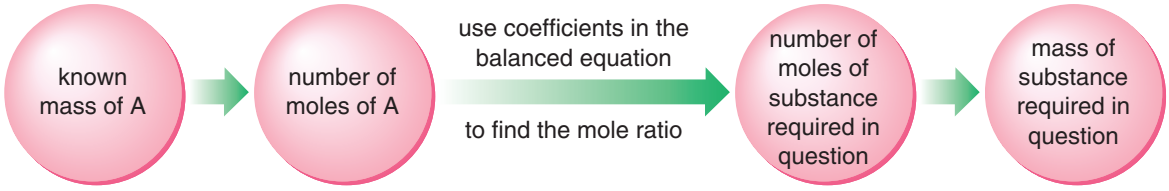
8 The empirical formula of a compound gives the simplest whole number ratio of atoms or ions present in the compound.

9 The following diagram summarizes the steps for determining the empirical formula of a compound.



Have you mastered?

10 The following diagram summarizes the steps for finding out the reacting masses of substances.



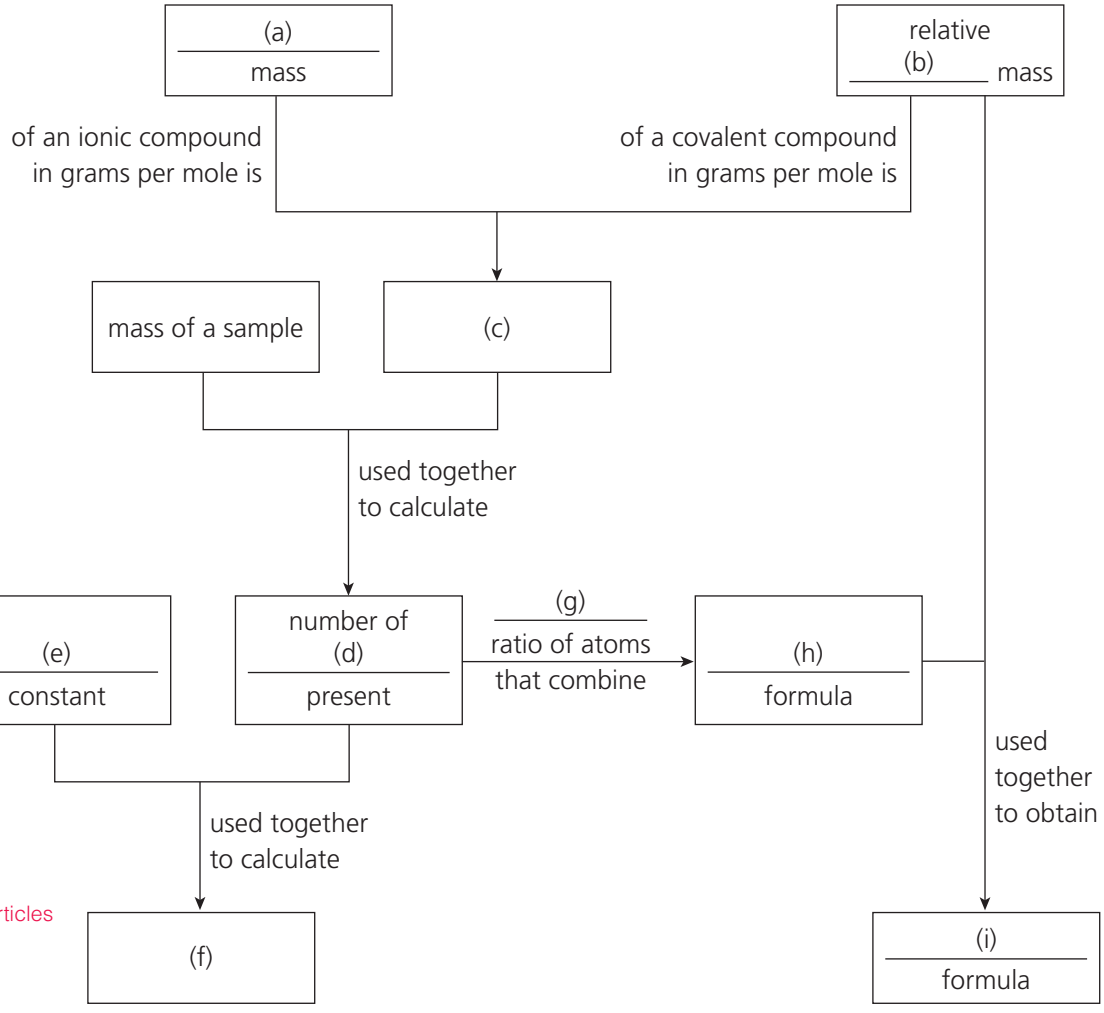
11 $\text{Percentage yield} = \dfrac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) formula
b) molecular
c) molar mass
d) moles
e) Avogadro
f) number of particles
g) mole
h) empirical
i) molecular

2 a) Number of moles of carbon atoms = $\frac{\text{mass of carbon}}{\text{molar mass of carbon}}$
$= \frac{0.321\text{ g}}{12.0\text{ g mol}^{-1}}$
$= 0.0268\text{ mol}$

2 Carbon occurs in a wide range of compounds and is essential to living systems.

In the sixteenth century, a large deposit of graphite was discovered in the Lake District of UK.

People at the time thought that the graphite was a form of lead.

Nowadays, graphite is used in pencils but it is still referred to as 'pencil lead'.

A student decided to investigate the number of carbon atoms in a 'pencil lead'. He found that the mass of the 'pencil lead' was 0.321 g.

a) Calculate the amount, in mole, of carbon atoms in the student's pencil lead.

Assume that the 'pencil lead' is pure graphite.

b) Calculate the number of carbon atoms in the student's 'pencil lead'.

(Relative atomic mass: C = 12.0; Avogadro constant = $6.02 \times 10^{23}\text{ mol}^{-1}$)

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2009, 1(c))

2 b) Number of carbon atoms
= number of moles of carbon atoms x Avogadro constant
$= 0.0268\text{ mol} \times 6.02 \times 10^{23}\text{ mol}^{-1}$
$= 1.61 \times 10^{22}$

3 Iron is an essential part of the human diet. Iron(II) sulphate is sometimes added to white bread flour to provide some of the iron in a person's diet.



a) The formula of iron(II) sulphate is $FeSO_4$. Calculate the formula mass of $FeSO_4$.

b) What is the mass of one mole of iron(II) sulphate?
151.9 g

c) What mass of iron(II) sulphate would be needed to provide 28.0 g of iron?
76.3 g; refer to page T41.
(Relative atomic masses: O = 16.0, S = 32.1, Fe = 55.8)

(AQA GCSE (Higher Tier), Chemistry, Unit 2, Jan. 2009, 1)

3 a) Formula mass of $FeSO_4$ = 55.8 + 32.1 + 4 x 16.0
= 151.9

Unit Exercise

Part II Multiple choice questions

4 One mole of carbon monoxide and one mole of carbon dioxide have the same

A mass.
B number of atoms.
(C) number of molecules.
D number of electrons.

One mole of carbon monoxide and one mole of carbon dioxide have the same number of molecules (6.02×10^{23} molecules).

5 Which of the following contains the greatest number of hydrogen atoms?
Refer to page T41 for explanation.

A 2 moles of water, H_2O
(B) 1.5 moles of ammonia, NH_3
C 1 mole of hydrogen gas, H_2
D 0.5 mole of methane, CH_4

(Edexcel Advanced Subsidiary GCE, Chemistry, Unit 1, Jan. 2009, 9)

6 How many formula units are there in 41.5 g of calcium carbonate?

(Relative atomic masses: C = 12.0, O = 16.0, Ca = 40.1; Avogadro constant = $6.02 \times 10^{23}\text{ mol}^{-1}$)

(A) 2.50×10^{23}
B 2.00×10^{23}
C 1.50×10^{23}
D 1.00×10^{23}

Number of moles of $CaCO_3$ = $\frac{\text{mass of } CaCO_3}{\text{molar mass of } CaCO_3}$
$= \frac{41.5\text{ g}}{100.1\text{ g mol}^{-1}}$
$= 0.415\text{ mol}$

Number of formula units of $CaCO_3$
= number of moles of $CaCO_3$ x Avogadro constant
$= 0.415\text{ mol} \times 6.02 \times 10^{23}\text{ mol}^{-1}$
$= 2.50 \times 10^{23}$

7 17.1 g of aluminium sulphate, $Al_2(SO_4)_3$, were dissolved in water.

Calculate the number of sulphate ions, SO_4^{2-}, present in the solution formed.
Refer to page T42 for explanation.
(The molar mass of $Al_2(SO_4)_3$ is 342.3 g mol^{-1} and the Avogadro constant is 6.02×10^{23} mol^{-1}.)

A 3.0×10^{21}
B 1.0×10^{22}
C 3.0×10^{22}
(D) 9.0×10^{22}

(Edexcel Advanced Subsidiary GCE, Chemistry, Unit 1, Jun. 2009, 5)

8 Which of the following samples of gases contains the greatest number of molecules?
Refer to page T42 for explanation.
(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0, S = 32.1)

(A) 10 g of CH_4
B 10 g of CO
C 10 g of NO
D 10 g of SO_2

9 A certain oxide of manganese contains 49.5% of manganese by mass. What is the empirical formula of this oxide?

(Relative atomic masses: O = 16.0, Mn = 54.9)

A MnO
B MnO_2
C Mn_2O_3
(D) Mn_2O_7

(HKCEE, Paper 2, 2010, 33)

10 Titanium (Ti) is a metal. 2.66 g of a sample of titanium powder are heated in excess oxygen until the metal is completely oxidized. The mass of the oxide formed is 4.44 g. Which of the following is the empirical formula of the oxide formed?

(Relative atomic masses: O = 16.0, Ti = 47.9)

A TiO
B Ti_2O_3
C Ti_3O_4
(D) TiO_2

(HKDSE, Paper 1A, 2013, 13)

11 A sample of $Na_2CO_3 \bullet xH_2O(s)$ of mass 134.0 g contains 81.0 g of water of crystallization. What is the value of x?
Refer to page T42 for explanation.
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0)

A 7
B 8
(C) 9
D 10

12 Decomposition of solid $KClO_3$ gives KCl and O_2 as the only products. What is the mass of O_2 produced when 49.0 g of $KClO_3$ undergo complete decomposition?
Refer to page T43 for explanation.
(Relative atomic masses: O = 16.0, Cl = 35.5, K = 39.1)

A 14.4 g
(B) 19.2 g
C 24.0 g
D 28.4 g

13 A sample of paint pigment of mass 1.50 g is dissolved and the lead(II) ions in it are separated by precipitation as lead(II) sulphate ($PbSO_4$). The precipitate has a mass of 0.0849 g. What is the percentage by mass of lead in the pigment?
Refer to page T43 for explanation.
(Relative atomic masses: O = 16.0, S = 32.1, Pb = 207.2)

(A) 3.87%
B 5.66%
C 5.80%
D 12.7%

14 Tin(IV) oxide, SnO_2, can be reduced by heating with carbon:

$$SnO_2(s) + 2C(s) \longrightarrow Sn(s) + 2CO(g)$$

332 g of the oxide are heated with 60.0 g of carbon. What is the mass of tin obtained?
Refer to page T43 for explanation.
(Relative atomic masses: C = 12.0, O = 16.0, Sn = 118.7)

A 135 g
B 178 g
C 237 g
(D) 261 g

15 Methanol (CH_3OH) can be obtained by the reaction between carbon monoxide and hydrogen:

$$CO(g) + 2H_2(g) \longrightarrow CH_3OH(g)$$

Excess hydrogen is allowed to react with 84.0 g of carbon monoxide. 81.6 g of methanol are obtained. What is the percentage yield of methanol?

Refer to page T44 for explanation.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

A 39.2%
B 77.5%
C 85.0%
D 97.1%

16 Which of the following statements concerning 1 mole of calcium ions are correct?

(1) It contains 6.02×10^{23} calcium ions.
(2) It contains the same number of particles as 2 moles of sodium ions.
(3) It contains the same number of particles as 12.0 g of $^{12}_{6}C$.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(2) 2 moles of sodium ions contain $2 \times 6.02 \times 10^{23}$ sodium ions.

17 b) ii) Percentage by mass of P in $(NH_4)_3PO_4$

$$= \frac{\text{number of atoms of P in formula} \times \text{relative atomic mass of P}}{\text{formula mass of } (NH_4)_3PO_4} \times 100\%$$

$$= \frac{31.0}{149.0} \times 100\%$$

$$= 20.8\%$$

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

17 A student uses the internet to find information about some salts.

Look at the table. It shows the information the student finds.

Name of salt	Formula of salt	Ions present in salt
Ammonium phosphate	$(NH_4)_3PO_4$	ammonium and phosphate
Barium sulphate	$BaSO_4$	barium and sulphate
Lead(II) nitrate	$Pb(NO_3)_2$	lead(II) and nitrate
Potassium iodide	KI	potassium and iodide
Potassium nitrate	KNO_3	potassium and nitrate

a) How many atoms are there in the formula for lead(II) nitrate? 9

b) i) What is the formula mass of ammonium phosphate? Formula mass of $(NH_4)_3PO_4$
$= 3 \times (14.0 + 4 \times 1.0) + 31.0 + 4 \times 16.0$
$= 149.0$

ii) What is the percentage by mass of phosphorus in ammonium phosphate?

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, P = 31.0)

c) A solution of silver nitrate is mixed with a solution of potassium iodide. A yellow precipitate of silver iodide, AgI, is made. Potassium nitrate solution is also made. Write an equation for this reaction.
$AgNO_3(aq) + KI(aq) \longrightarrow AgI(s) + KNO_3(aq)$

d) The student decides to make barium sulphate by reacting barium chloride with ammonium sulphate.

He starts with 2.00 g of barium chloride.

He does not get a 100% yield of barium sulphate.

He predicts he should make 2.24 g of barium sulphate. He actually makes 1.68 g of barium sulphate.

What is his percentage yield of barium sulphate?

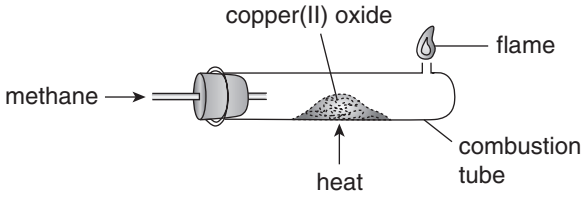
(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2011, 2)

Percentage yield of barium sulphate

$$= \frac{1.68\text{ g}}{2.24\text{ g}} \times 100\%$$

$$= 75.0\%$$

18 An experiment was done on the reaction of copper(II) oxide (CuO) with methane (CH_4).



a) The equation for this reaction is shown below.

$$4CuO(s) + CH_4(g) \longrightarrow 4Cu(s) + 2H_2O(g) + CO_2(g)$$

Use information from the equation to explain why the water and carbon dioxide produced escape from the combustion tube. Because they are gases.

b) i) Calculate the formula mass of copper(II) oxide (CuO). 79.5

Refer to page T44.

(Relative atomic masses: O = 16.0, Cu = 63.5)

ii) Calculate the percentage by mass of copper in copper(II) oxide.
79.9%

iii) Calculate the mass of copper that could be made from 4.00 g of copper(II) oxide. 3.19 g

(AQA GCSE (Higher Tier), Chemistry, Unit C2, Jun. 2011, 2(a)–(b))

19 Tin mining was common practice on Dartmoor in pre-Roman times. Most of the tin extracted was mixed with copper to produce bronze.

Refer to page T45.

a) A bronze-age shield found on Dartmoor contained 2.08 kg of tin (Sn).

Calculate the number of tin atoms in this bronze shield.
1.05×10^{25}

(Relative atomic mass: Sn = 118.7; Avogadro constant = 6.02×10^{23} mol^{-1})

b) Tin ore, known as cassiterite, contains an oxide of tin. This oxide contains 78.8% tin by mass. Calculate the empirical formula of this oxide.

(Relative atomic mass: O = 16.0)

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2010, 1(c)–(d))

20 Some students investigated magnesium oxide.

Refer to page T45.

a) Magnesium oxide has the formula MgO.

i) Calculate the formula mass of magnesium oxide.
40.3

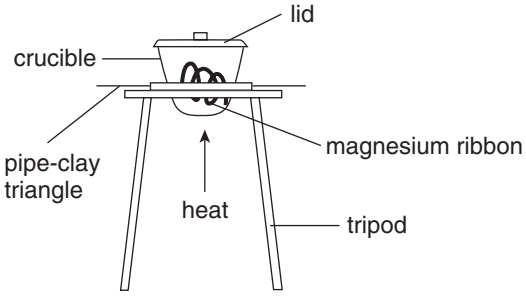
(Relative atomic masses: O = 16, Mg = 24.3)

ii) Calculate the percentage by mass of magnesium in magnesium oxide.
60.3%

iii) Calculate the mass of magnesium needed to make 25.0 g of magnesium oxide.
15.1 g

b) The students calculated that if they used 0.120 g of magnesium they should make 0.200 g of magnesium oxide.

They did this experiment to find out if this was correct.



- The students weighed 0.120 g of magnesium ribbon into a crucible.
- They heated the magnesium ribbon.
- They lifted the lid of the crucible slightly from time to time to allow air into the crucible.
- The students tried to avoid lifting the lid too much in case some of the magnesium oxide escaped.
- When all of the magnesium appeared to have reacted, the students weighed the magnesium oxide produced.

The results of the experiment are shown below.

Mass of magnesium used (g)	0.120
Mass of magnesium oxide produced (g)	0.180

i) The mass of magnesium oxide produced was lower than the students had calculated. They thought that this was caused by experimental errors.

Suggest TWO experimental errors that the students had made.

ii) The students only did the experiment once.

Give TWO reasons why they should have repeated the experiment.

(AQA GCSE (Higher Tier), Chemistry, Unit 2, Jan. 2012, 2)

21 Sodium burns in oxygen to give a pale yellow solid X.

a) i) 1.73 g of sodium react with 1.20 g of oxygen.

Calculate the empirical formula of X.
Refer to page T46.
(Relative atomic masses: O = 16.0, Na = 23.0)

ii) The molar mass of X is 78.0 g mol^{-1}. Give the chemical formula of X.
Refer to page T46.

iii) Write the equation, including state symbols, for the reaction of sodium with oxygen to produce X.
$2Na(s) + O_2(g) \longrightarrow Na_2O_2(s)$

iv) Calculate the number of oxygen molecules that react with 1.73 g of sodium.
2.26×10^{22}; refer to page T46.
(The Avogadro constant = 6.02×10^{23} mol^{-1})

b) If sodium is burnt in air, compound X is NOT the only product. Suggest why this is so.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 25)

22 Sulfamic acid is a white solid used by plumbers as a limescale remover.

Sulfamic acid contains 14.42% by mass of nitrogen, 3.09% of hydrogen and 33.06% of sulphur. The remainder is oxygen.
Refer to page T47.

a) Calculate the empirical formula of sulfamic acid.

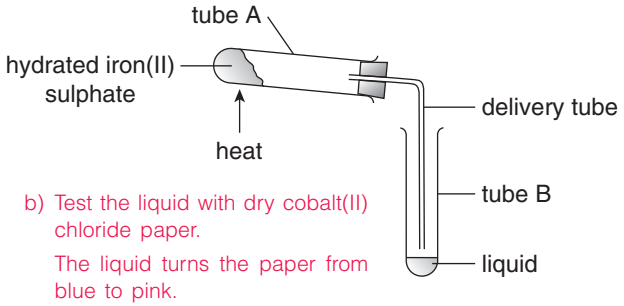
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, S = 32.1)

21 b) Sodium may react with nitrogen in the air to form a nitride. / Sodium may form a different oxide (e.g. Na_2O).

b) The molar mass of sulfamic acid is 97.1 g mol^{-1}. Use this information to deduce the molecular formula of sulfamic acid.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2010, 17(a))

23 The set-up shown below was used to collect the liquid given off when hydrated iron(II) sulphate was heated strongly for some time.



b) Test the liquid with dry cobalt(II) chloride paper.
The liquid turns the paper from blue to pink.

a) Explain why tube A was clamped in a slanting position. To prevent the condensed water from running back to the tube and crack the hot glass.

b) Suggest a test to show that the liquid collected in tube B was water.

c) Why is it desirable to keep the free end of the delivery tube above the liquid collected in tube B?
To prevent 'sucking back' of the liquid.

d) The formula for hydrated iron(II) sulphate is $FeSO_4 \bullet xH_2O$. In the above experiment, 30.6 g of the sulphate produced 13.9 g of water. Calculate the value of *x*.
7; refer to page T47.
(Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Fe = 55.8)

24 Glauber's salt is a form of hydrated sodium sulphate that contains 44.1% by mass of sodium sulphate. Hydrated sodium sulphate can be represented by the formula $Na_2SO_4 \bullet xH_2O$ where *x* is an integer. Calculate the value of *x*.
10; refer to page T48.
(Relative atomic masses: H = 1.0, O = 16.0, Na = 23.0, S = 32.1)

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2010, 2(e))

Unit Exercise

25 M_2O is an oxide of metal M. Upon heating, M_2O decomposes to give M and oxygen only.

a) Suggest a method for testing oxygen, and state the expected observation.

b) In an experiment, 3.48 g of M_2O completely decompose to give 3.24 g of M. Calculate the relative atomic mass of M.

c) Explain whether M can react with dilute hydrochloric acid.

(HKCEE, Paper 1, 2010, 4)

26 Iron is extracted by heating iron(III) oxide with carbon.

During this reaction the carbon takes the oxygen away from the iron(III) oxide.

a) What do we call a reaction in which a metal oxide loses oxygen?

Circle the correct answer.

electrolysis extraction oxidation (reduction)

b) The formula of one oxide of iron is Fe_2O_3.

What mass of iron is contained in 160 g of Fe_2O_3?

(Relative atomic masses: O = 16.0, Fe = 55.8)

Circle the correct answer.

48 g 56 g 64 g (112 g)

c) Put numbers in the boxes to balance the equation for the formation of iron from iron(III) oxide.

$2Fe_2O_3 +$ [3] $C \longrightarrow$ [4] $Fe +$ [3] CO_2

d) Other metals are also extracted by heating their oxides with carbon.

Put ticks '✔' in the boxes next to TWO metals that are extracted in this way.

Copper	✔
Magnesium	
Potassium	
Sodium	
Zinc	✔

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 2, Jan. 2012, 2)

27 Titanium(IV) oxide is used as a white pigment in some paints. The pigment can be made as shown in the following equation.

$$TiCl_4(l) + 2H_2O(l) \longrightarrow TiO_2(s) + 4HCl(aq)$$

In an experiment 165 g of $TiCl_4$ were added to an excess of water. Calculate the maximum mass of TiO_2 formed in this experiment.

69.4 g; refer to page T48.

(Relative atomic masses: O = 16.0, Cl = 35.5, Ti = 47.9)

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2009, 3(b)(iii))

28 In the UK over 1 million tonnes of sulphuric acid are manufactured each year.

a) The first stage in the manufacture of sulphuric acid is to make sulphur dioxide.

One way of making sulphur dioxide is by heating iron sulphide, FeS_2, in air to make solid iron(III) oxide, Fe_2O_3, and sulphur dioxide, SO_2.

Add numbers before the formulae on the right hand side to balance the equation for this reaction.

Fill in the missing state symbols.

$4FeS_2(s) + 11O_2(g)$
$\longrightarrow$ 2 Fe_2O_3(s) + 8 SO_2(g)

b) Another way of making sulphur dioxide is by burning sulphur in air.

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$

Calculate the maximum mass of sulphur dioxide that could be made by burning one tonne of sulphur.

(Relative atomic masses: O = 16.0, S = 32.1)

(Edexcel GCSE (Higher Tier), Chemistry, Unit C3, Jun. 2011, 7(b)–(c))

S(s) + O_2(g) $\longrightarrow$ SO_2(g)
1 tonne ? tonnes

Molar mass of S = 32.1 g mol^{-1}

Molar mass of SO_2 = (32.1 + 2 x 16.0) g mol^{-1}
= 64.1 g mol^{-1}

According to the equation, 1 mole of S can produce 1 mole of SO_2.

∴ 32.1 g of S can produce 64.1 g of SO_2.

Mass of SO_2 = 1 tonne x $\frac{64.1\text{ g}}{32.1\text{ g}}$
= 2 tonnes

∴ the maximum mass of SO_2 that could be made is 2 tonnes.

29 At a high temperature, calcium oxide reacts with carbon to form calcium carbide, CaC_2.

$$CaO(s) + 3C(s) \longrightarrow CaC_2(s) + CO(g)$$

Calcium oxide reacts with carbon to make 128 g of calcium carbide. Calculate the minimum mass, in g, of carbon required.
72.0 g; refer to page T48.
(Relative atomic masses: C = 12.0, Ca = 40.1)

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2, Nov. 2008, 9(a))

30 Sodium nitrate decomposes when heated, as shown by the equation:

$$2NaNO_3(s) \longrightarrow 2NaNO_2(s) + O_2(g)$$

A 1.70 g sample of sodium nitrate was completely decomposed to sodium nitrite and oxygen.

Calculate the mass of sodium nitrite formed.
1.38 g; refer to page T49.
(Relative atomic masses: N = 14.0, O = 16.0, Na = 23.0)

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 1C, May 2011, 10(f))

31 Aluminium can be produced by heating aluminium oxide with magnesium.

a) Write a chemical equation for the reaction involved.
$Al_2O_3(s) + 3Mg(s) \longrightarrow 3MgO(s) + 2Al(s)$
b) What mass of aluminium oxide is required to produce 16.2 g of aluminium?
30.6 g; refer to page T49.
c) What mass of magnesium is required to react with the mass of aluminium oxide calculated in (b)?
21.9 g; refer to page T50.

(Relative atomic masses: O = 16.0, Mg = 24.3, Al = 27.0)

32 In the early 1990s, sodium azide, NaN_3, was used to inflate airbags in cars if there was a collision.



The bag inflates because the solid sodium azide decomposes explosively to produce sufficient nitrogen gas to fill the nylon bag in less than 0.1 seconds.
Refer to page T50.

a) The equation for the explosive reaction in which sodium azide produces nitrogen is

$$2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$$

A car manufacturer requires enough sodium azide to make 2 500 moles of nitrogen, N_2. Calculate the number of moles of sodium azide, NaN_3, required to make this amount of nitrogen.
1 670 mol

b) 3.00 moles of nitrogen gas are required to fill a different airbag.

Calculate the minimum mass of solid sodium azide, in grams, required to produce the nitrogen gas required to fill this airbag.
130 g
(Relative atomic masses: N = 14.0, Na = 23.0)

(Edexcel GCSE (Higher Tier), Chemistry, Unit C3, May 2010, 9(b)–(c))

33 Upon strong heating, silver oxide decomposes to give silver and oxygen. An impure 8.00 g sample of silver oxide gave 6.52 g of silver upon strong heating. Calculate the percentage by mass of silver oxide in the sample.
87.5%; refer to page T50.
(Relative atomic masses: O = 16.0, Ag = 107.9)

34 Manganese(IV) oxide reacts with concentrated hydrochloric acid according to the following equation:

$$MnO_2(s) + 4HCl(aq) \longrightarrow MnCl_2(aq) + Cl_2(g) + 2H_2O(l)$$

217 g of manganese(IV) oxide were treated with hydrochloric acid containing 274 g of hydrogen chloride. Determine
Refer to page T51.

a) the limiting reactant;

b) the mass of chlorine produced; and
133 g
c) the mass of excess reagent left after the reaction.
54 g

(Relative atomic masses: H = 1.0, O = 16.0, Cl = 35.5, Mn = 54.9)